



# Wheat Production

Compiled by

**Dr Rorisang Patose and Mr Moses Ncala**

([PatoseR@arc.agric.za](mailto:PatoseR@arc.agric.za) & [NcalaM@arc.agric.za](mailto:NcalaM@arc.agric.za))

Agricultural Research Council – Small Grain Institute



**agriculture, land reform  
& rural development**

Department:  
Agriculture, Land Reform and Rural Development  
**REPUBLIC OF SOUTH AFRICA**



**forestry, fisheries  
& the environment**

Department:  
Forestry, Fisheries and the Environment  
**REPUBLIC OF SOUTH AFRICA**

**giz** Deutsche Gesellschaft  
für Internationale  
Zusammenarbeit (GIZ) GmbH

On behalf of:



Federal Ministry  
for the Environment, Nature Conservation  
and Nuclear Safety

of the Federal Republic of Germany

**ARC • LNR**  
Excellence in Research and Development

## Table of Contents

|       |                                                                            |     |
|-------|----------------------------------------------------------------------------|-----|
| 1     | INTRODUCTION                                                               | 188 |
| 2     | IMPACT OF CLIMATE CHANGE ON WHEAT PRODUCTION                               | 191 |
| 2.1   | EFFECTS OF CLIMATE CHANGE ON WHEAT DEVELOPMENT,<br>YIELD AND GRAIN QUALITY | 192 |
| 2.1.1 | Germination and emergence                                                  | 192 |
| 2.1.2 | Tillering                                                                  | 192 |
| 2.1.3 | Stem elongation                                                            | 192 |
| 2.1.4 | Heading/flowering                                                          | 193 |
| 2.1.5 | Grain-fill / ripening                                                      | 193 |
| 2.2   | SOIL HEALTH                                                                | 196 |
| 2.3   | CROP PESTS AND PATHOGENS                                                   | 196 |
| 2.3.1 | Insect pests                                                               | 196 |
| 2.3.2 | Diseases                                                                   | 201 |
| 2.3.3 | Weeds                                                                      | 204 |
| 3     | INTERVENTIONS ON THE IMPACT OF CLIMATE CHANGE                              | 208 |
| 3.1   | SOIL HEALTH                                                                | 208 |
| 3.1.1 | Organic matter                                                             | 209 |
| 3.1.2 | Conservation agriculture                                                   | 212 |
| 3.2   | CROP PESTS AND PATHOGENS                                                   | 218 |
| 3.2.1 | Insect pests                                                               | 218 |
| 3.2.2 | Diseases                                                                   | 219 |
| 3.2.3 | Weeds                                                                      | 219 |
| 4     | CONCLUSION                                                                 | 220 |
| 5     | REFERENCES AND RESOURCES                                                   | 221 |
|       | LIST OF FIGURES                                                            | 224 |
|       | LIST OF TABLES                                                             | 225 |

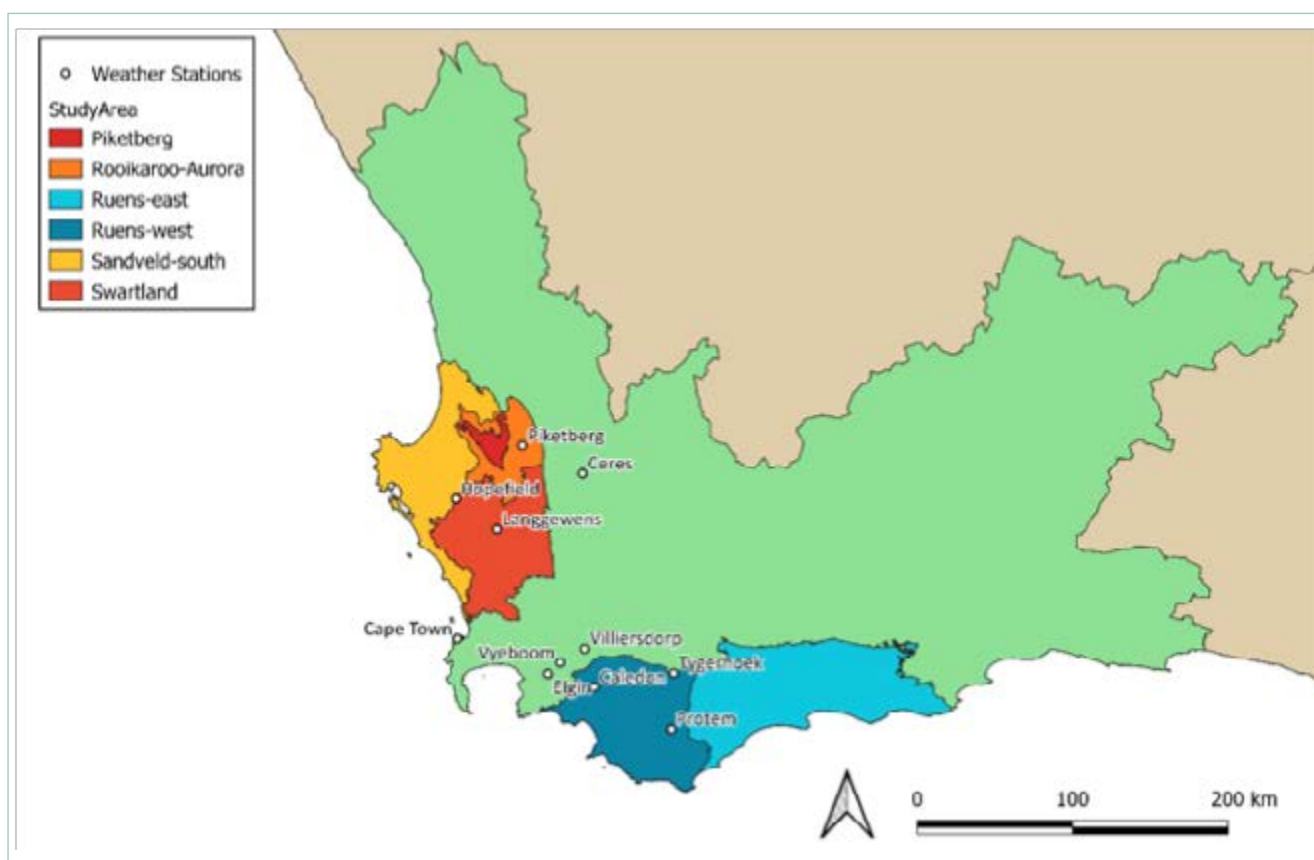
## 1 INTRODUCTION

### Objective

1. Understand the importance of rainfall and irrigation in wheat production.

The main wheat-producing provinces in South Africa (SA) are the Western Cape (winter rainfall) ([Figure 1](#)), Free State (summer rainfall) and Northern Cape (irrigation) ([Figure 2](#)). Wheat is also produced under irrigation in Mpumalanga and mainly under irrigation in Eastern Cape, Limpopo and North West ([Figure 2](#), and [Table](#)

[1](#)). The annual wheat production in South Africa ranges from 1.5-3 million tonnes, with productivity levels of 2-2.5 t ha<sup>-1</sup> under dryland and at least 5 t ha<sup>-1</sup> under irrigation. However, wheat production has been decreasing in recent years.



**Figure 1** Map showing both dryland wheat production areas in winter rainfall areas of Western Cape Province.

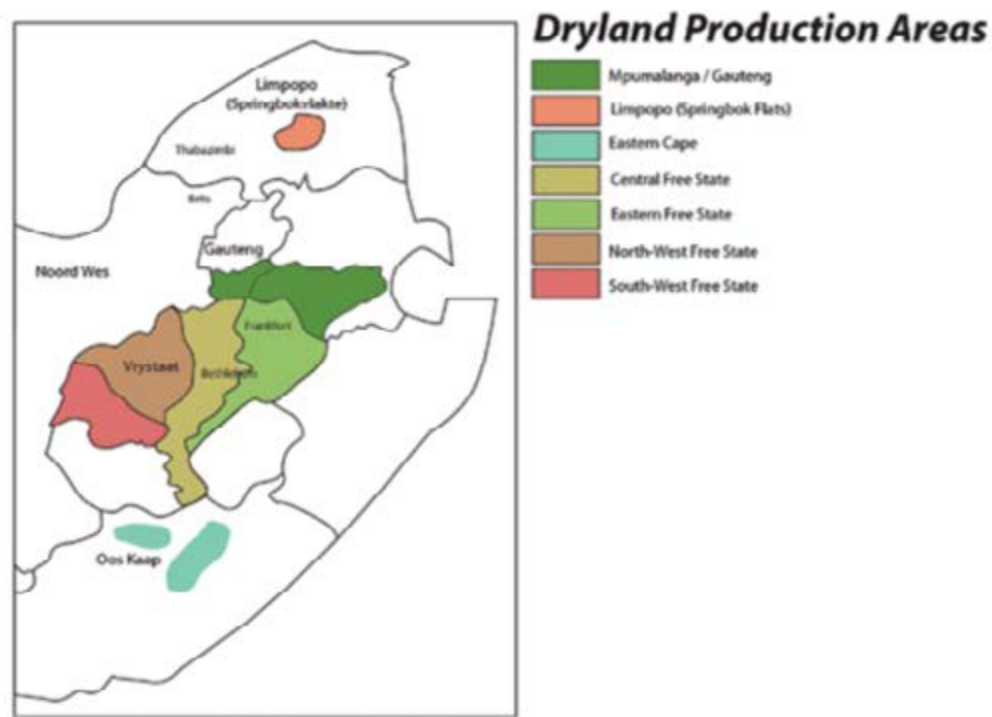
*Source: Personal communication Simone Theron, ARC-NRE.*

Table 1 shows that, under dryland conditions, the number of hectares planted with wheat has decreased in the three provinces over the last three years, with yields fluctuating.

South Africa is a net importer of wheat, importing a large amount each year to meet local demand because it does not produce enough.

**Table 1 Wheat production overview over three seasons.**  
*Source: Southern African Grain Laboratory, 2018 & 2019.*

| Province     | Type of production | 2016/2017        |                             | 2017/2018        |                             | 2018/2019        |                             |
|--------------|--------------------|------------------|-----------------------------|------------------|-----------------------------|------------------|-----------------------------|
|              |                    | Hectares planted | Yield (t ha <sup>-1</sup> ) | Hectares planted | Yield (t ha <sup>-1</sup> ) | Hectares planted | Yield (t ha <sup>-1</sup> ) |
| Eastern Cape | Dryland            | 700              | 2.43                        | 600              | 2.33                        | 400              | 2.5                         |
|              | Irrigation         | 1500             | 6.2                         | 1300             | 6.23                        | 1250             | 7.78                        |
|              | Total              | 2200             |                             | 1900             |                             | 1650             |                             |
| Limpopo      | Dryland            | 850              | 2.35                        | 1000             | 3.6                         | 1000             | 2.5                         |
|              | Irrigation         | 16150            | 6.3                         | 19000            | 6.76                        | 19000            | 6.61                        |
|              | Total              | 17000            |                             | 20000            |                             | 20000            |                             |
| North West   | Dryland            | 80               | 2.5                         | 80               | 2.5                         | -                | -                           |
|              | Irrigation         | 11920            | 5.8                         | 13420            | 6.22                        | 1200             | 6.5                         |
|              | Total              | 12000            |                             | 15500            |                             | 1200             |                             |



### Irrigation Areas of South Africa



Figure 2 Map showing both dryland and irrigated wheat production areas in summer rainfall areas of SA.  
Source: ARC-SG, Bethlehem.

## 2 IMPACT OF CLIMATE CHANGE ON WHEAT PRODUCTION

### Objectives

1. Understand the effects of climate change on wheat development, yield and grain quality.
2. Understand what impact climate change and variability have on soil health.
3. Learn about crop pests and pathogens.

Climate change causes environmental changes that have a negative impact on crop production around the world. Temperature changes, rainfall variability, changes in atmospheric gas composition, and changes in solar radiation are all examples of environmental changes that affect crop, pest, pathogen, and weed relationships, as well as soil health (Myers et al, 2017). The net effect of [climate change](#) on crop yields and quality is determined by the interactions of all of these factors, and as [adaptation](#) and [mitigation](#) measures are implemented, the cost of inputs will rise.

Agricultural, Forestry, and Other Land Use (AFOLU) activities contributed for roughly 13% of CO<sub>2</sub>, 44% of methane (CH<sub>4</sub>), and 81% of nitrous oxide (N<sub>2</sub>O) emissions from human activities globally between 2007 and 2016, accounting for 23% (12.0 2.9 GtCO<sub>2</sub> eq yr<sup>-1</sup>) of total net anthropogenic GHG emissions (medium confidence). During the period 2007-2016, land's natural response to human-induced environmental change resulted in a net sink of roughly 11.2 Gt CO<sub>2</sub> yr<sup>-1</sup> (equal to 29% of global CO<sub>2</sub> emissions) (medium confidence); the sink's persistence is unknown due to climate change (high confidence) (IPCC, 2019). If emissions from pre- and post-production activities in the global food system are included, total net anthropogenic GHG emissions are projected to be 21-37% (medium confidence) (IPCC, 2019).

Climate change risk is determined by the rate of warming as well as population, consumption, production, technological progress, and land management trends (high confidence). Water shortages in drylands, land degradation, and food insecurity are all problems that come with greater demand for food, feed, and water, more resource-intensive consumption and production, and restricted technological advancements in agriculture yields (high confidence) (IPCC, 2019).

Globally, average temperatures have risen, resulting in higher rates of [evaporation](#) and drier soil conditions (Girvetz and Zganjar, 2014; Girvetz et al, 2019). While temperatures are expected to rise as a result of climate change, rainfall projections differ between global climate models (GCM) (Flato et al., 2013). The magnitude of these changes was expected to vary greatly between regions (CSIRO and Bureau of Meteorology, 2015). In the next 30–80 years, atmospheric CO<sub>2</sub> levels are expected to exceed 550 parts per million. Protein, iron, and zinc content in many food crops grown below 550 ppm are reduced by 3–17%, potentially causing protein, iron, and zinc deficiency in people.

Since 1975, temperatures in Africa have risen at a rate of about 0.03 degree Celsius per year (Girvetz et al, 2019). And the new normal for temperature is higher than any other time in

history. Much of Africa is drying out, according to precipitation patterns (Hartmann et al, 2013). In comparison to other parts of Southern Africa, such as Zambia and Zimbabwe, South Africa has received more rain.

Climate change-induced temperature increases, according to Easterling et al. (2007), are expected to reduce wheat production in developing countries by 20-30%. The region's low productivity (2t/ha) is primarily due to abiotic (drought and heat) and biotic (yellow rust, stem rust, septoria, and fusarium) stresses that are intensifying and occurring more frequently as a result of climate change (Tadesse et al, 2018).

Climate change's impact on crop production must be understood in order to adapt and mitigate it. The following section discusses the impact of climate change on wheat production.

Climate change affects the health of the soil on which crops grow, as well as the distribution and occurrence of pests, pathogens, and weeds, all of which have an impact on wheat development, yield, and quality. Wheat development, yield and grain quality, soil health, and crop pests and pathogens will all be covered in this section (insect pests, diseases and weeds).

## **2.1 EFFECTS OF CLIMATE CHANGE ON WHEAT DEVELOPMENT, YIELD AND GRAIN QUALITY**

### **2.1.1 Germination and emergence**

Seed germination and seedling emergence are influenced by a variety of factors, including available moisture and soil temperature. To germinate, seed must consume a significant amount of water in relation to its dry weight. Wheat seed must have a minimum water

content of 35–40% by weight in order to germinate. Seed that germinates at a lower moisture content may make it through the first stages, but it will not perform to its genetic potential. Excess moisture, on the other hand, causes germination to be delayed or prevented due to a lack of oxygen. Photosynthesis requires a constant supply of water after emergence.

Soil temperature also plays a significant role in the rate at which germination proceeds. Germination can take place at temperatures ranging from 4°C to 37°C, but the best temperatures for germination are 12°C to 25°C. Temperature has an impact on the rate of imbibition. Temperatures below the optimum result in lower germination and longer germination periods, whereas temperatures above the optimum reduce or prevent germination by denaturing cellular protein and killing the seed.

### **2.1.2 Tillering**

A temperature increase of 10–25°C increases the total number of leaves and tillers, whereas a temperature increase of more than 25°C reduces leaf development. Temperature fluctuations cause increased leaf senescence, which can reduce grain yield by 50%. High temperatures cause crop water loss due to increased respiration rates, resulting in lower biomass production and, as a result, lower yields. Lower leaf area occurs as a result of water stress.

### **2.1.3 Stem elongation**

Water stress results in fewer nodes, lower stem dry weights, and a decrease in height. The increase in average temperature during the growing season typically causes plants to use more energy for respiration and less for growth.



#### 2.1.4 Heading/flowering

Winter and spring wheat differ in vernalization. Winter wheat cultivars will not flower until after a sufficiently extended period of cold or low temperatures to enable flowering. However, spring wheat varieties will flower as soon as possible. Because wheat has such a wide range of vernalization requirements, the choice of "winter" and "spring" variants is according to the climatic conditions of the specific location. An increase in temperatures, will therefore decrease the vernalization period and can negatively affect flowering. Water deficit at flowering also limits the viability of pollen, the receptivity of its stigma, and seed development and fertilisation. When higher temperatures occur during the seed formation period, it will be a shorter time period, hence fewer wheat grains will be initiated and that will result in lower yields.

#### 2.1.5 Grain-fill / ripening

Air temperatures above approximately 30°C are generally associated with lower yields in rain-fed crops like dryland wheat (Carlson, 1990; Schlenker & Roberts, 2009). High temperatures can limit yields by accelerating crop development (Asseng et al, 2014; Butler & Huybers, 2015) particularly by shortening the grain filling stages, meaning the time when sugars from the leaves is transported to the seed storage place is limited, so even if there have been sufficient seeds initiated, they may not be filled. Water deficits can also induce direct damage of plant cells (Sánchez et al, 2014). Climatic variability in terms of rainfall (drought, &/or excessive rain) and terminal heat (i.e. high temperature during grain filling stage) have a severe negative impact on wheat production. For example, if wheat is planted too late, high temperatures at

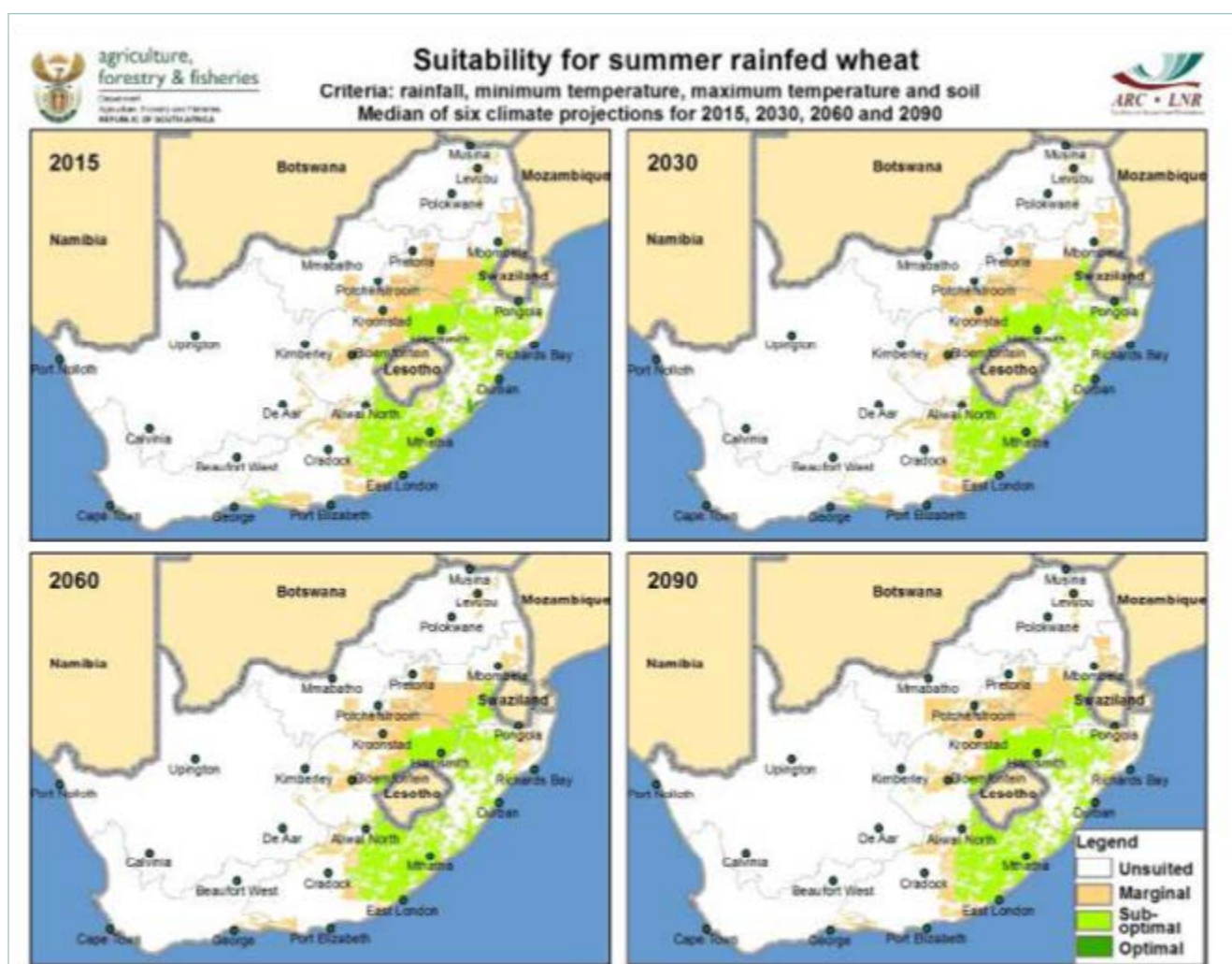
the end of season hastens maturity and reduces grain yield (commonly known as terminal heat stress). Wheat then matures earlier than normal leading to a lower yield. High temperature and terminal drought negatively affect wheat biomass and grain yield. Under drought stress dry weight accumulation is lower, resulting in significantly lower crop yields. The increasing carbon dioxide (CO<sub>2</sub>) content in the atmosphere increases crop performance by increasing rates of photosynthesis and water use efficiency but can substantially decrease grain quality of wheat. Elevated CO<sub>2</sub> decreases grain protein, Zinc (Zn) and Iron (Fe) and their responses to elevated are also reduced by water stress and nitrogen (N) (Myers et al, 2014; Al-Hadeethi et al., 2019).

The GCM outputs were used together with certain crop characteristics to make a prediction of the probably shift in the production areas of wheat across South Africa. Crop specialists assisted in the development of environmental compatibility criteria. These criteria tables were used to create a temporal series of maps [baseline (2015) near-future (2030), mid-century (206) and near-end century (2090)] depicting probable movements in wheat production areas (Weepener, et al., 2014). As different crops are affected in different ways by climate change, production areas of most crops could decrease (examples include maize, soybean, sorghum, sunflower, potato and Smuts finger grass), while the production area of some crops could increase (sugarcane, groundnut and cotton), depending on their interaction with the changing temperatures and rainfall. Wheat, however, mostly remain unchanged. Nevertheless, producers will have to adapt to the changing climate to be in a better position to provide for the demand for food. The following series of maps show the projected changes in the



wheat production areas by using 2015 period as a baseline for current production. Some of the areas west of Potchestroom, NW where spring wheat is planted in June the summer rainfall areas, appear to be more marginal towards the middle of the century (Figure 3). For the inland irrigated wheat production areas, there is little change, except by mid-century, the many

areas in North West province move from being optimal to being sub-optimal (Figure 4). For the eastern provinces irrigated areas, the optimal areas are also shifted to higher altitudes and further from the coast, where the crop can still to benefit from the cooler temperatures (Figure 4).



**Figure 3** Projection of the possible shifts in the wheat production areas for the summer rainfall areas where spring wheat is planted in the easterly provinces.

Source: Weepenaar et al., 2014.

## Climate-Smart Agriculture – Training Manual

### WHEAT PRODUCTION

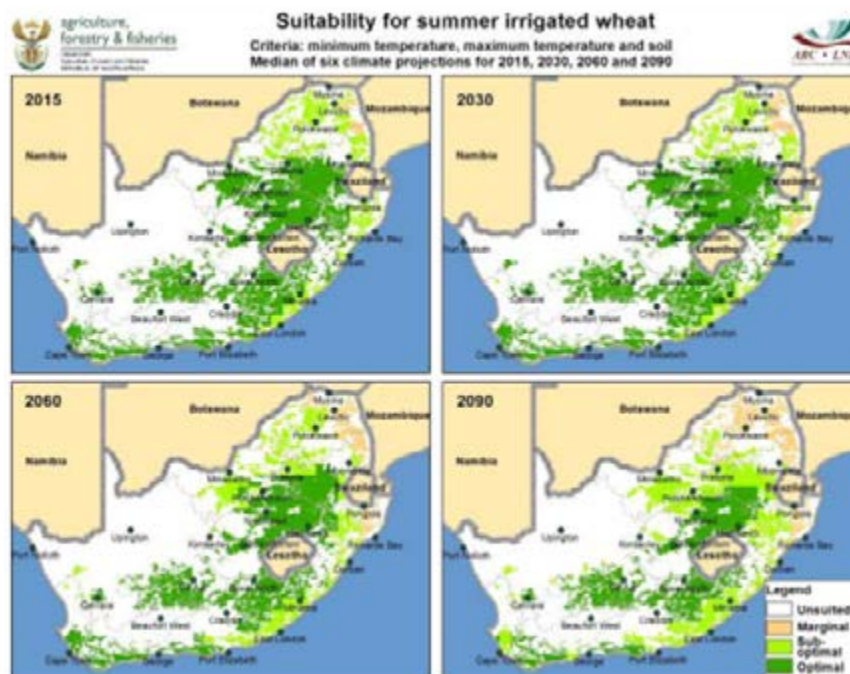


Figure 4 Projection of the possible shifts in the wheat production areas under irrigation in the summer rainfall areas.

Source: Weepenaar et al., 2014.

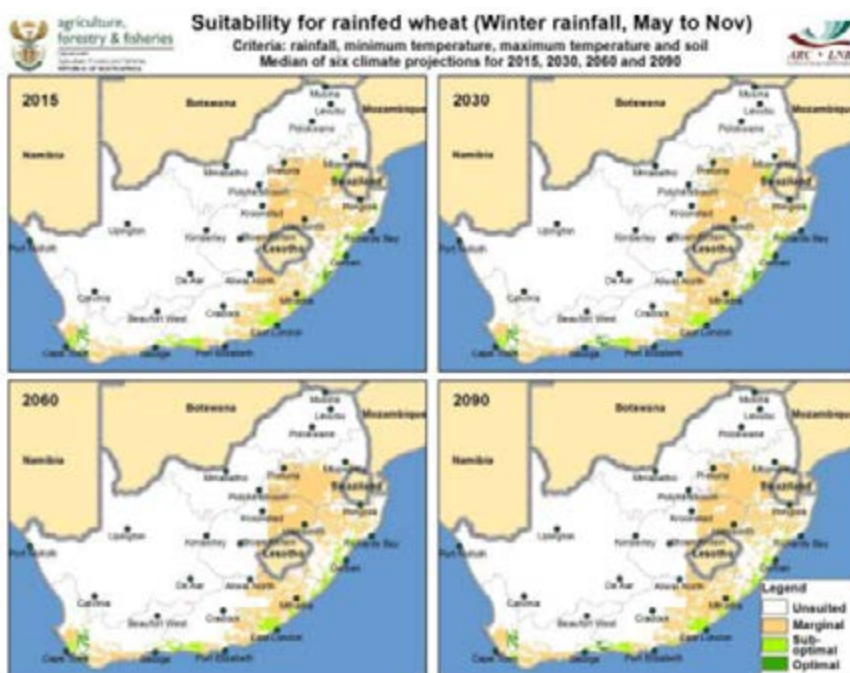


Figure 5 Projection of the possible shifts in the wheat production areas for Western Cape winter rainfall area.

Source: Weepenaar et al., 2014.

Many of the irrigated areas in northern part of Limpopo province will also shift from being sub-optimal to being marginal for wheat production (Figure 4). Little change is expected in the production areas in the Western Cape winter rainfall area where much of the South African wheat is produced under rainfed or dryland conditions (Figure 5). This is indeed good news for the farmers, however, they will need to use climate-smart agricultural techniques, as it is expected that there will be more frequent droughts (Theron et al., 2021 & 2022).

## 2.2 SOIL HEALTH

[Climate change](#) and variability have a large impact on crop growth and yields because of their impact on soil health and crop varieties' ability to adapt to changing climate and weather patterns. A healthy soil for wheat development must have sufficient essential [nutrients](#), water, a suitable temperature, oxygen (air), and the ability to support plant roots. Heavy rain causes runoff and soil erosion, resulting in significant losses of soil organic matter and poor crop production (GrainSA, 2015).

Soil erosion causes changes in soil structure, as well as leaching and nutrient loss. The loss of salts and nutrient cations during leaching increases salinization in areas where there is net upward water movement due to increased evapotranspiration or decreased rainfall or irrigation water supply (Chandra et al, 2013). Because the production, transport, and application of [fertiliser](#) emits more carbon dioxide, soil fertilisation used to improve essential nutrients does not typically generate a net sink for carbon. Fertilisers cause less water, air, and life in the soil (GrainSA, 2015).

## 2.3 CROP PESTS AND PATHOGENS

Plant-pest interactions are changing as a result of climate change. Plants that grow in waterlogged soil are more susceptible to viruses and soil-borne diseases. The distribution of insect pest pests is influenced by temperature. Plant pathogens' ecology can be altered by climate extremes, and higher soil temperatures can encourage fungal growth.

In general, rising CO<sub>2</sub> levels and temperatures increase pest occurrence, severity of damage, and pest infestation winter survival, as well as the spread of disease and pest-carrying organisms. Increased rainfall may result in leaf wetness and humid conditions, which can aid in the development and spread of a variety of diseases and pests.

### 2.3.1 Insect pests

Aphids, mites, false wire worms, Bagrada bugs, African bollworms, leaf miner flies, and false armyworms are among the wheat insect pests. Their significance, however, is determined by the wheat production area and the climatic conditions in which they can thrive. As a result, it is difficult to rank these pests in terms of overall importance across the country, though three insects or insect groups can be highlighted as being important in almost every production area of the country. The three will be discussed in greater depth.

#### 2.3.1.1 Aphids

Aphids are the most common insect pests of wheat and other small grains, and they can be found in nearly every wheat-growing region. This group contains about five prominent species, the importance of which varies by region depending on weather conditions.

The first is the Russian wheat aphid (RWA) (see [Figure 6](#)), which thrives in dry, water-stressed environments. They are particularly important in the Free State and Western Cape's dryland environments. With the rise in dryland wheat production, this pest may become more important.

RWA is the most damaging aphid on wheat due to its feeding damage resulting in considerable yield losses. Managing RWA also proved to be a challenge. Resistance in crops is an effective management tool for insect crop pests.

A drawback of this management option is that the insect can overcome the resistance in the crop, which will result in new insect biotypes able to damage the resistant crop plant. This proved to be the case with RWA resistant wheat cultivars.

RWA is the most damaging aphid on wheat because of its feeding damage, which results in significant yield losses. Managing RWA has also proven to be difficult. Crop resistance is an effective tool for controlling insect crop pests. One disadvantage of this management option is that the insect can overcome crop resistance, resulting in new insect biotypes capable of damaging the resistant crop plant. This was demonstrated by RWA resistant wheat cultivars. South Africa currently has five RWA biotypes. RWASA1 has been present in South African wheat production areas since 1978 and has been able to overcome the *Dn3* resistant gene in wheat (see [Table 2](#)).



**Figure 6 Russian wheat aphid.**  
*Source: Dr Goddy Prinsloo.*



In 2005, RWASA2 was recorded in the Eastern Free State. This RWA biotype was able to overcome the *Dn1*, *Dn2*, *Dn3*, *Dn8*, *Dn9*, and *Dn y* resistant genes in wheat (see [Table 2](#)). RWASA3 was recorded in the Eastern Free State in 2009 and has a *Dn4* resistant gene added to its virulence profile. RWASA4 was recorded in the Eastern Free State in 2011 and has a *Dn5* resistant gene added to its virulence profile (see

[Table 2](#)). RWASA5 was recorded in 2018 and has the *Dn6* and *Dnx* genes added to its virulence profile. Currently, RWASA5 is the most virulent RWA biotype. RWA biotypes are found in a complex, primarily in the Eastern Free State. Only RWASA1 is found in wheat production areas in the rest of the country (Western and Central Free State, Northern Cape, and Western Cape).

**Table 2 Resistance(R)/Susceptibility (S) of wheat germplasm with resistant genes against five South African biotypes (RWASA1-RWASA5).**

Source: SA Grain, March 2019.

| Resistant gene | RWA biotype |        |        |        |        |
|----------------|-------------|--------|--------|--------|--------|
|                | RWASA1      | RWASA2 | RWASA3 | RWASA4 | RWASA5 |
| <i>Dn1</i>     | R           | S      | S      | S      | S      |
| <i>Dn2</i>     | R           | S      | S      | S      | S      |
| <i>Dn3</i>     | S           | S      | S      | S      | S      |
| <i>Dn4</i>     | R           | R      | S      | S      | S      |
| <i>Dn5</i>     | R           | R      | R      | S      | S      |
| <i>Dn7</i>     | R           | R      | R      | R      | R      |
| <i>Dn8</i>     | R           | S      | S      | S      | S      |
| <i>Dn9</i>     | R           | S      | S      | S      | S      |
| <i>Dnx2006</i> | R           | R      | R      | R      | S      |
| <i>Dny2006</i> | R           | S      | S      | S      | S      |

The second most important group of aphids includes the oat aphid (shown in [Figure 7](#)), rose grain aphid (shown in [Figure 8](#)), and brown ear aphid (shown in [Figure 9](#)). They thrive in the Western Cape's humid winter rainfall, and they are common in all irrigation areas. They occur sporadically in the Free State during years when

there is a lot of rain in the fall. The humid climate is ideal for these aphids. These aphids are not as dangerous as the Russian wheat aphid, but feeding damage of 12% to 15% may occur during heavy infestations. They are, however, known to be effective virus vectors, particularly for the Barley Yellow Dwarf Virus (BYDV).



**Figure 7 Oat aphid.**  
*Source: Dr Goddy Prinsloo.*



**Figure 8 Rose grain aphid.**  
*Source: Dr Goddy Prinsloo.*

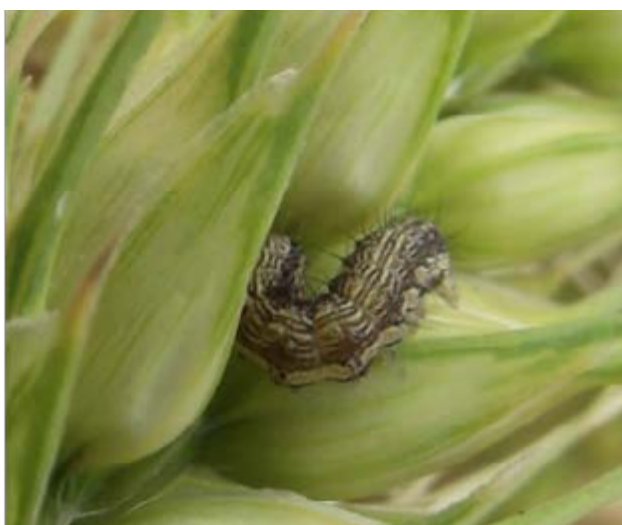


**Figure 9 Brown ear aphid.**  
*Source: Dr Goddy Prinsloo.*

#### 2.3.1.2 African bollworm

The African bollworm, pictured in [Figure 10](#), ranks second among insect pests of wheat and other small grains because it is a voracious feeder found in all production areas, albeit not on a regular basis. Although this insect is

potentially more damaging in irrigated wheat, significant infestations occur sporadically in dryland conditions in the Western Cape and Free State. Larvae have been observed moving into the heads at an early stage, where they feed on the kernels, causing them to be damaged.



**Figure 10 African bollworm and the damage it causes.**  
*Source: Dr Goddy Prinsloo.*



### 2.3.1.3 Leaf miner flies

The leaf miner fly first appeared as a wheat pest in irrigated wheat fields in the Prieska and Douglas areas in 2000. This is a local indigenous insect that began infesting wheat and barley for unknown reasons. Initially, the pest spread north along the major rivers to Vaalharts and Bloemhof. Since then, it has gradually spread to Lichtenburg and the Brits area. It first appeared on dryland wheat in the Western Cape in 2016, where one to two life cycles were completed early in the wheat growing season. Despite their presence in KwaZulu-Natal production areas, they are not a problem, owing to natural enemy action.

### 2.3.2 Diseases

The latitudinal range of pathogens shifts as temperatures change. Extreme weather events can destabilise agricultural systems, weakening crop defenses and allowing pathogens to establish themselves in niches.

### 2.3.2.1 Rusts

Diseases are more likely to occur and spread as temperatures and rainfall rise. When there is an abundance of free water, for example, wheat stem and leaf rusts increase with rising temperatures. Stripe rust infection, on the other hand, is most common in cool weather (below 15°C). As a result, stripe rust is more common in cooler wheat-growing regions (e.g., Eastern Free State), while stem rust and leaf rust are more common in wheat-growing regions with mild winter temperatures (e.g. Western Cape).

When conditions are favorable, susceptible cultivars are planted, cultural practices are altered, and these three factors combine, rusts can cause severe losses. The environmental conditions required by the three rust diseases, namely leaf rust (shown in [Figure 11](#)), stem rust (shown in [Figure 12](#)) and stripe rust (shown in [Figure 13](#)) are summarized in [Table 3](#).



**Figure 11 Leaf rust.**  
*Source: Dr T. Terefe.*



**Figure 12 Stem rust.**  
*Source: Dr T. Terefe.*



**Figure 13 Stripe rust.**  
*Source: Dr T. Terefe.*

**Table 3 Environmental conditions required for the wheat rusts.**

Source: Roelfs et al., 1992.

| Stage        | Temperature (°C) |         |              | Light     | Free water |
|--------------|------------------|---------|--------------|-----------|------------|
|              | Minimum          | Optimum | Maximum      |           |            |
| Leaf rust    |                  |         |              |           |            |
| Germination  | 2                | 20      | 30           | Low       | Essential  |
| Germling     | 5                | 15-20   | 30           | Low       | Essential  |
| Appressorium | -                | 15-20   | -            | None      | Essential  |
| Penetration  | 10               | 20      | 30           | No effect | Essential  |
| Growth       | 2                | 25      | 35           | High      | None       |
| Sporulation  | 10               | 25      | 35           | High      | None       |
| Stem rust    |                  |         |              |           |            |
| Germination  | 2                | 15-24   | 30           | Low       | Essential  |
| Germling     | -                | 20      | -            | Low       | Essential  |
| Appressorium | -                | 16-27   | -            | None      | Essential  |
| Penetration  | 15               | 29      | 35           | High      | Essential  |
| Growth       | 5                | 30      | 40           | High      | None       |
| Sporulation  | 15               | 30      | 40           | High      | None       |
| Stripe rust  |                  |         |              |           |            |
| Germination  | 9                | 9-13    | 23           | Low       | Essential  |
| Germling     | -                | 10-15   | -            | Low       | Essential  |
| Appressorium | -                | -       | (not formed) | None      | Essential  |
| Penetration  | 2                | 8-13    | 23           | Low       | Essential  |
| Growth       | 3                | 12-15   | -            | High      | None       |
| Sporulation  | 5                | 12-15   | -            | High      | None       |

### 2.3.2.2 Barley Yellow Dwarf Virus (BYDV)

Virus infections such as BYDV, such as shown in [Figure 14](#), are common in some irrigation areas, resulting in a 33% yield loss. The occurrence and incidence of barley yellow dwarf virus are influenced by environmental factors. The disease thrives in cool temperatures (10° - 18°C)

combined with moist conditions. Because this virus is only transmitted by aphids, aphid control is critical early in the growing season, when wheat is most susceptible to BYDV. The oat aphid, rose-grain aphid, and English grain aphid are the most common aphids on small grains in South Africa that can transmit BYDV.

These aphids are all present on wheat that has been irrigated, as well as in Kwazulu-Natal and the Northwest province, where BYDV infections have occurred on a regular basis in recent years. Climate change will have an impact on the population dynamics of these vectors, as well as the prevalence of BYDV.

### 2.3.3 Weeds

Weed control in any small grain production system can be difficult, especially with the emergence of herbicide-resistant weeds. Weed infestations in wheat have been shown to reduce wheat yield by up to 33%. Several post-emergence herbicides have been shown to no longer control grassweeds in recent seasons. Many farmers were forced to implement an integrated weed management system (IWM) and are now focusing on pre-emergence herbicide control strategies.

Extreme weather events can destabilise agricultural systems by undermining crop

defenses and creating niches for weeds to thrive. Because of their diverse gene pool, weeds are better adapted to rising  $\text{CO}_2$  concentrations and rising temperatures as a result of climate change. Weeds can be classified as C3 or C4 weeds, and their responses to higher  $\text{CO}_2$  levels and temperatures differ.

This will have an impact on crop-weed competition dynamics. Furthermore, climate change factors may impact the efficacy of many herbicides, making weed management a significant challenge for sustainable crop production. Chemical control methods are heavily used in wheat production. Herbicides accounted for nearly 25% of total [pesticide](#) use worldwide in 2011. (Grube et al., 2011). Herbicides have become the primary tool for weed management due to their ease of use, higher efficacy, and, most importantly, lower control costs due to labor and time savings (McErlich and Boydston, 2013).



**Figure 14 Barley yellow dwarf virus infected wheat leaves.**

**Source:** <https://www.gardeningknowhow.com/edible/grains/barley/barley-yellow-dwarf-virus.htm>.

The efficiency of herbicide absorption into the plant is determined by several factors. Temperature, moisture, relative humidity, and solar radiation are some of the factors to consider. These factors have an impact on a plant's physiologic state and herbicide susceptibility. Interactions between these factors make determining their effects on herbicide performance even more difficult. Weeds, which have more genetic diversity and physiological plasticity than crops,

respond quickly to resource changes and have a higher likelihood of adapting and reproducing in different habitats. Weeds with  $C_3$  or  $C_4$  photosynthetic pathways have different responses to climate change.

Some examples of  $C_3$  weeds are: Wild buckwheat (*Fallopia convolvulus*, [Figure 15](#) and [Figure 16](#)), Lambsquarters (*Chenopodium album*, [Figure 17](#) and [Figure 18](#)) and Wild Oats (*Avena fatua*, [Figure 19](#) and [Figure 20](#)).



**Figure 15 Wild buckwheat seedling.**  
*Source: Hestia Nienaber.*



**Figure 16 Wild buckwheat in an oats field.**  
*Source: Hestia Nienaber.*





**Figure 17** Young *Chenopodium* plant.  
*Source: Hestia Nienaber.*



**Figure 18** *Chenopodium* seeds.  
*Source: Hestia Nienaber.*



**Figure 19** *Avena* plant.  
*Source: Hestia Nienaber.*



**Figure 20 Avena spikelets.**  
*Source: Hestia Nienaber.*

All of the weeds mentioned above can quickly disperse and establish themselves in new areas, posing a threat to [ecosystem](#) composition and integrity. Many weeds are expected to expand their geographical range and cause greater crop productivity losses as warmer winters make higher latitudes more conducive to plant growth and higher temperatures make lower latitudes less habitable (Patterson, 1995; Parmesan and Yohe, 2003).

An increase in  $\text{CO}_2$  levels in the atmosphere will almost certainly have a significant impact on weed biology. Herbicide performance on weeds will be affected as a result. The most noticeable effect of increased atmospheric  $\text{CO}_2$  levels is a reduction in stomatal conductance, which in some plants can increase by up to 50%. (Bunce, 1993). Reduced stomatal conductance, in turn, can affect herbicide efficacy, both foliar and soil-applied. Furthermore, as  $\text{CO}_2$  levels rise,

leaf thickness rises and the number of open stomata falls, reducing the amount of foliar-applied herbicide that is directly absorbed into the plants and protecting weeds from post-emergence herbicide damage.

The uptake of herbicides applied to the soil is further reduced by reduced stomatal conductance (Bunce & Ziska, 2000; Ziska, 2008). Furthermore, at high  $\text{CO}_2$  levels, an increase in net photosynthetic rates, particularly in C3 weeds, could result in rapid seedling growth. Because the seedling stage is the most vulnerable for effective weed control, the timing of post-emergence herbicide application may need to be adjusted. Lambsquarters is an example of a C3 weed that has demonstrated increased tolerance to glyphosate as a result of increased growth and biomass at higher  $\text{CO}_2$  levels (Ziska et al., 1999).

### 3 INTERVENTIONS ON THE IMPACT OF CLIMATE CHANGE

#### Objectives

1. Understand the importance of soil health.
2. Learn about crop pests and pathogens.

To achieve the highest possible yield, farmers can use high-quality seed and planting materials of well-adapted varieties; a diverse range of crop species and varieties grown in associations, intercrops, or rotations; pest control through integrated pest management; and conservation agriculture and sustainable mechanization to maintain healthy soils and manage water efficiently (FAO, 2011).

It's also crucial to understand the appropriate operations to perform on a small grain field, the appropriate time to perform them, the appropriate method to perform them, as well as why the operations are performed, why they are performed at a specific time, and why they are performed in a specific manner. Climate-smart small grain production: what, when, and how.

#### 3.1 SOIL HEALTH

Higher yields are difficult to achieve in soils depleted of essential [nutrients](#), water, oxygen (air), and plant root support. Poor crop production results from soil degradation, which is usually caused by soil tillage by increasing water run-off and soil erosion, as well as significant losses of soil organic matter (GrainSA, 2015). Improving the soil's quality is therefore critical.

A healthy agricultural soil, shown in [Figure 21](#), is one that can support the production of food and fibre at a level and quality sufficient to meet human needs, as well as the continued delivery of other [ecosystem services](#) critical to human quality of life and biodiversity conservation (Kibblewhite et al, 2008).



**Figure 21** Healthy loose soil with good structure versus unhealthy cloddy soil with poor structure.  
**Source:** <https://www.rolawn.co.uk/soil-structure>.

A healthy soil is alive and provides plants with essential nutrients, water, oxygen (air), and root support. Healthy soil, contains many living organisms. It is deep, loose, full of air and water and it is easy to work on.

A healthy soil has a good structure that creates air pockets allowing water to infiltrate and move deep into the soil. Healthy soils act as giant moisture sponges, which is critical during times of drought and flooding. To accomplish this, the soil requires a continuous supply of organic matter and the accumulation of organic matter, and conservation agriculture is the solution. Conservation agriculture increases [resilience](#) by improving soil quality.

### 3.1.1 Organic matter

Organic matter in the soil consists of three distinctly different parts: living organisms, fresh residues and well decomposed residues. The living organisms in the organic matter includes a wide variety of microorganisms, such as bacteria, viruses, fungi, protozoa and [algae](#). It also includes plant roots, insects, earthworms, and larger animals such as moles and rabbits. The living portion represents about 15% of the total soil organic matter.

#### 3.1.1.1 Why organic matter is so important?

Living organisms in soil help to control insect pests, weeds, and plant diseases, form beneficial symbiotic relationships with plant roots, recycle plant [nutrients](#) from soil organic matter and minerals back to roots, and improve soil structure. Earthworms (shown in [Figure 22](#)) are an example of an organism that helps to increase the amount of air and water that enters the soil. They convert organic matter, such as leaves and grass, into nutrients that plants can use. When they eat, they leave behind castings that are a very valuable type of [fertiliser](#). Earthworms are essentially free farm workforce.

Essentially all soil properties are positively influenced or modified by organic matter. As organic matter increases, soils tend to be less compact and have more space for air passage and water storage. On the other hand, as soil organic matter declines, it becomes increasingly difficult to grow plants due to increased problems with fertility, water availability, compaction, erosion, parasites, diseases, and insects.



**Figure 22 Earthworms in healthy soil.**

**Source:** <https://grist.org/food/feed-your-soil-and-the-rest-will-follow>.



To maintain yields in the face of organic matter depletion, ever higher levels of inputs, fertilisers, irrigation water, pesticides, and machinery are required. However, with proper organic matter management, the soil can support a good crop without the need for costly fixes.

The addition of organic matter to the soil increases the soil's water holding capacity. Because water is the primary medium for moving nutrients into plants as a result of water uptake, more water in the soil means more nutrient uptake by crops.

The rise in soil temperature due to deficiency of soil moisture has an effect on microbial activity and nutrient processing, both of which are important for plant use in biomass and grain production. Soil moisture and temperature, in general, control microbial activity in soil. Deviations from the optimum ranges of soil moisture (water field capacity) and soil temperature, which vary for different microbial communities in soil, can alter microbial activity. During a drought, changes in soil temperature can affect organic matter decomposition and increase carbon dioxide emissions. In addition,

additional mineral [nitrogen](#), primarily in the form of nitrate, will be released into the soil system during this process. This change in soil environment has an impact on the stability of soil organic matter and, as a result, the soil biological system.

The most significant effect in cropland is the excessive release of nitrate, which may not be utilised by crops due to a lack of moisture available for the plant to absorb nutrients. This shift in biological and chemical processes during the growing season influences many other relationships that are essential for crop performance, both quantitatively and qualitatively, by changing activities that are important to nutrient cycling, such as enzymatic activities, changes in soil chemical concentrations, and so on.

The well-decomposed organic material in the soil is known as humus (see [Figure 23](#)). Microorganisms convert simple sugars or liquid carbon exuded from plant roots to humus. Some essential [nutrients](#) are stored in humus and released slowly to plants.



**Figure 23** Wheat seedlings growing on healthy soil with well decomposed humus.

**Source:** <https://regenerationinternational.org/2015/10/08/emerging-land-use-practices-rapidly-increase-soil-organic-matter>.



Humus can also act as a barrier to certain potentially harmful chemicals, preventing them from harming plants. Humus in soils has an average age of more than 1,000 years due to its stability and complexity. Although the already decomposed humus is not a food source for organisms, its small size and chemical properties make it an important component of a healthy soil.

### 3.1.1.2 How to increase soil organic matter

1. Use crop residues more effectively and find new sources of residues to add to soils. New residues can include those grown on the farm, such as cover crops, or those available from various local sources.
2. Try to use a number of different types of materials - crop residues, manures, composts, cover crops, leaves, etc. It is important to provide varied residue sources to help develop and maintain a diverse group of soil organisms.

3. Although use of organic materials from off farm can be a good source for building soil organic matter and adding nutrients, some farmers overload their fields with excess nutrients by excess imports of organic materials. Crop residues (including cover crops) as well as on-farm-derived animal manures and composts help to supply organic materials and cycle nutrients without a buildup of excessive levels of nutrients.

4. Implement practices that decrease the loss of organic matter from soils because of accelerated decomposition or erosion.

All practices that aid in the increase of organic matter levels either add more organic materials or reduce the rate of organic matter loss from soils. Furthermore, practices that increase organic matter will usually benefit beneficial organisms while stressing pests (see [Table 4](#)).

**Table 4 Effects of different management practices on gains and losses of organic matter, beneficial organisms and pests.**

| Management Practice                                                           | Gains Increase? | Losses Decrease? | Enhance beneficials (EB), Stress Pests (SP) |
|-------------------------------------------------------------------------------|-----------------|------------------|---------------------------------------------|
| Add materials (manures, composts, other organic materials) from off the field | yes             | no               | EB, SP                                      |
| Better utilise crop residues and mulches                                      | yes             | no               | EB                                          |
| Include high-residue-producing crops in rotation                              | yes             | no               | EB, SP                                      |
| Include sod crops (grass/legume forages) in rotation                          | yes             | yes              | EB, SP                                      |
| Grow cover crops                                                              | yes             | yes              | EB, SP                                      |
| Reduce tillage intensity                                                      | yes/no*         | yes              | EB                                          |
| Use conservation practices to reduce erosion                                  | yes/no*         | yes              | EB                                          |

\*Practice may increase crop yields, resulting in more residue.

Practices that combine the two may be particularly beneficial. Practices that reduce organic matter losses either slow the rate of decomposition or reduce erosion. Soil erosion must be controlled in order to keep organic matter-rich topsoil in place. In addition, organic matter added to a soil must either match or exceed the rate of loss by decomposition. These additions can come from off-field manures and composts, crop residues and mulches that remain following harvest, or cover crops.

### 3.1.2 Conservation agriculture

Conservation agriculture (CA) aims to achieve sustainable and profitable agriculture through the application of the three CA principles of conserving, improving, and making more efficient use of soil, water, and biological resources (plants, animals, insects, and microbes). The three CA principles are:

1. Continuous minimum mechanical soil disturbance;
2. Permanent organic soil cover; and
3. Diversification of crop species grown in sequences and/or associations.

Conservation farming holds tremendous potential for farms of all sizes, but the most significant limitation is the initial lack of knowledge when a producer decides to switch from plough-based farming to CA.

#### 3.1.2.1 Minimum tillage

This is when the soil is not ploughed. The goal is to disturb the soil as little as possible. The seeds are planted directly into the mulch covered field using specialised no-till planters ([Figure 24](#)). The soil is disturbed only where seed and fertility amendments ([fertiliser](#), manure, compost) are to be placed.



**Figure 24 Minimum tillage planting.**

**Source:** [https://commons.wikimedia.org/wiki/File:Striptill\\_2011\\_09\\_07\\_1709.jpg](https://commons.wikimedia.org/wiki/File:Striptill_2011_09_07_1709.jpg).

#### 3.1.2.1.1 Advantages of minimum tillage

The benefits of disturbing the soil as little as possible are as follows:

1. It ensures minimum destruction of the soil structure.
2. It protects the soil from wind and water erosion.
3. It allows for slower mineralisation of organic matter, resulting in the accumulation of organic matter.
4. It has little impact on the life of soil organisms, which helps to improve soil structure.
5. It saves time, energy, and money because there is less ploughing and fertility amendments are only applied to planting areas.

#### 3.1.2.2 Mulching / Soil cover

The soil remains covered with either crop residue (Figure 25 and Figure 26), other types of mulch like cover crop (see Figure 27) or growing

plants on the soil at all times. In CA, crop residue is typically left on the field to cover the soil. Other types of mulch can be used to fill in the gaps between the planting rows. Mulching reduces soil erosion and soil temperature by at least 4°C, creating better conditions for soil organisms to thrive.

#### 3.1.2.2.1 Advantages of mulching

1. Properly managed mulching improves water infiltration resulting in a higher soil water content.
2. It helps in reducing direct raindrop impact and run-off in the field thus reducing soil erosion.
3. It reduces [evaporation](#) and conserves soil moisture.
4. It keeps soil moisture even and cool.
5. It aids in the suppression of weeds.
6. It provides food and a conducive environment for soil organisms, which are essential for biological processes and soil fertility.



**Figure 25** Soil covered with crop residue.

**Source:** <https://www.no-tillfarmer.com/articles/489-no-till-movement-in-us-continues-to-grow?v=preview>.



**Figure 26** Soil in between wheat rows covered with crop residue.

**Source:** [https://www.researchgate.net/figure/Effect-of-crop-establishment-straw-mulch-and N-application-on-grain-yield-t-ha-of-tbl3-299389081](https://www.researchgate.net/figure/Effect-of-crop-establishment-straw-mulch-and-N-application-on-grain-yield-t-ha-of-tbl3-299389081).



**Figure 27** Soil covered with cover crop between rows of wheat crop.

**Source:** <https://mail.sssup.it/~barberi/research.htm>.



### 3.1.2.3 Crop diversification, mixed cropping and crop rotation

#### 3.1.2.3.1 Crop diversification

Crop diversification is the practice of growing multiple varieties of crops from the same or different species in the same area. Crop diversification, as shown in [Figure 28](#), is one method of constructing a resilient agricultural system. Crop diversification is also one of the most environmentally friendly, feasible, cost-effective, and rational ways to reduce agricultural risks. Crop diversification also increases [resilience](#), or an [ecosystem](#)'s ability to return to its original productive state after being disturbed, by increasing both spatial and temporal biodiversity on the farm.

Farmers can also manage production risks by cultivating a variety of crop species. Crop diversification can be a viable strategy for increasing crop productivity in moisture-stressed, ecologically fragile agriculture systems.

All of these advantages of diversification contribute to a higher yield.

Because the goal is to mimic nature, as much diversity as possible is created. In the field, diversity ensures a natural balance. This includes things like cultivating a living soil, preventing weeds, conserving water, and reducing pest and disease attacks on crops.

Biodiversity on top of the soil equals biodiversity below the soil, which include the presence of living roots in the soil for the entire year. Maximum cover on top of the soil by plants either living or dead serve as armour to the soil just as our skin protects us from the sun and the rain. It keeps the soil cooler in summer and warmer in winter. This all leads to build-up of carbon in the soil, which is vital for farms' sustainability. For every 1% of added carbon to the soil, the water holding capacity of that soil doubles.



**Figure 28 Crop diversification by alternating crops at the same time.**

**Source:** <http://www.arc.agric.za/arc-sgi/Pages/Production%20Systems/Conservation-agriculture.aspx>.



### 3.1.2.3.2 Mixed cropping

Mixed cropping, also known as intercropping, entails planting multiple crops on the same piece of land at the same time. [Figure 29](#) shows wheat intercropped with clover, which also serves as a cover crop. Clover has many advantages in this intercropping because it improves soil quality and reduces the need for artificial [fertiliser](#). The incorporation of legume biomass into the soil after decomposition improves wheat [nitrogen](#) status and grain nitrogen content. It also provides nutrient dense forage for grazing animals after the harvesting of wheat. Intercropping also suppresses weeds. Single planted areas have more weeds because of the space

between the rows of plant. Intercropped areas form a canopy that quickly covers the soil, unlike single crop plants.

In this system food crops are mixed with soil enriching crops which:

- can fix nitrogen into the soil (legumes) and cycle plant [nutrients](#);
- grow quickly and provide a lot of above-ground (leaf) and below-ground (root) biomass;
- and improve soil biology, fertility, and structure both while growing and decomposing in the soil.



**Figure 29** Wheat mix cropped with clover.

**Source:** <https://www.country-guide.ca/crops/back-to-cover-crop-basics-with-red-clover>.

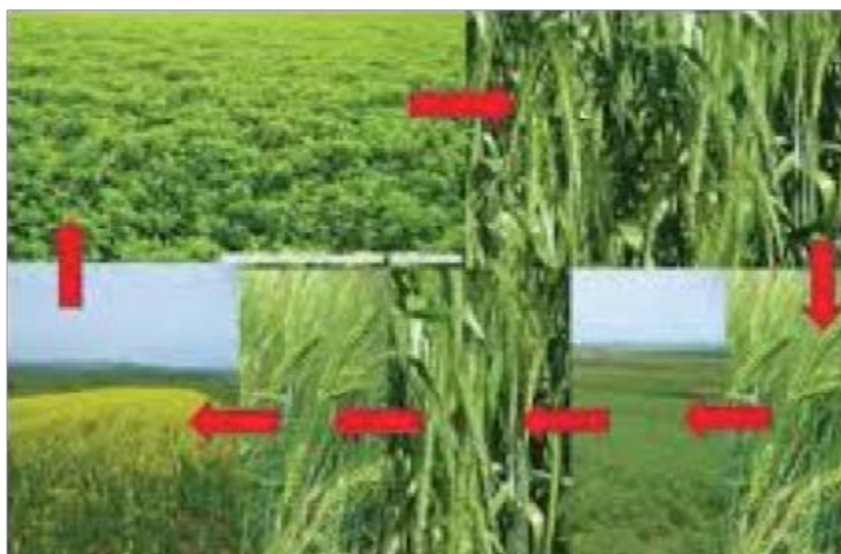
### 3.1.2.3.3 Crop rotation

Crop rotation, as depicted in [Figure 30](#), is when crops are planted in the same place at different times. Different crops are alternated on the same field from year to year. The aim is to alternate crop types as much as possible. By alternating grain crops, like wheat, with broadleaf crops such as canola, lupin or lucerne the opportunity of diversity in the field is created. Having at least three different crops is ideal. The best rotation is one that also provides fodder for livestock.

Crop rotation interrupts the disease cycle that can be destructive if the same crop is planted year after year. By incorporating legume crops such as lupin, vetch, peas, and legume pastures, one can fix nitrogen from the air for free. The nitrogen fixed by legume plants will decompose the following year, allowing farmers to reduce the amount of nitrogen fertiliser needed in the subsequent crop, thus saving money.

### 3.1.2.3.4 Advantages of crop diversification, mixed cropping and crop rotation

1. Soil fertility replenishment – Nitrogen fixing legumes add top dressing [fertiliser](#) to the soil.
2. Crops make better use of soil [nutrients](#). Because different crops have different feeding zones, they will not compete for nutrients. The use of different soil layers by different crops also aids in the prevention of hard pan.
3. It aids in disease and pest control because the introduction of a different crop disrupts the life cycles of these pests and diseases.
4. The soil structure benefits when it is occupied by roots from a variety of plants because:
  - The roots move the soil;
  - the roots form a network of living matter that dies and rots to form humus;



**Figure 30** Crop rotation, starting from top left, lucerne - wheat - barley - lupin - wheat - barley - canola - lucerne again.

**Source:** <https://www.grainsa.co.za/improve-your-wheat-yield-with-crop-rotation>.

- and when the roots die, they leave tunnels that improve porosity and drainage;
- roots secrete weak acids to dissolve minerals in the soil and then draw these minerals back up in solutions;
- roots also secrete a portion of their photosynthetic energy in the form of sugars that feed microbes, which in turn provide soil mineral nutrients to the roots.

## 3.2 CROP PESTS AND PATHOGENS

Out-of-season and in-season surveys and monitoring of insect pests, diseases, and weeds are conducted in order to make control recommendations. The use of integrated pest management practices is also encouraged.

Soil sampling is done prior to planting to identify soil-borne insect pests and pathogens, as well as weed seeds. Crop seeds are treated to keep insect pests and pathogens at bay. In the case of insect pests, traps are built during the growing season to collect data that can be used to alert farmers. Diseases such as rusts are monitored on an annual basis in order to detect new strains and recommend appropriate control. Regular field monitoring is critical because previously unknown diseases/pests may be introduced to a specific area as a result of [climate change](#). Monitoring will aid in the early detection and control of such diseases and pests.

Crop diversification can also control weeds effectively. Post-harvest crop protection surveys are also done for storage moths and weevils control especially for small scale farmers who store their produce after harvest. Use of improved post-harvest storage attributes to minimize the application of storage chemicals for insect pests and pathogens.

Integrated pest management (IPM) is also used, which entails using appropriate measures to discourage the development of pest populations and keep pesticides and other interventions to economically justified levels; reducing or minimising risks to human health and the environment; and disrupting the agricultural [ecosystem](#) as little as possible. The IPM manual can be found at <https://www.agriseta.co.za/downloads/LearningMaterial/116301LG.pdf>.

### 3.2.1 Insect pests

Biological measures can be used to control insect pests, here is a link to a manual that can be used: <https://agribook.co.za/inputs/biocontrol/#biocontrol>.

#### 3.2.1.1 Aphids

To prevent oat, rose grain, and brown ear aphid feeding damage, chemical control should be applied at the flag stage, when 25% of tillers have more than 10 aphids per tiller. Irrigated fields in areas where BYDV has previously been found should be treated as a precaution. Aphid migration is monitored using suction traps to provide early warning.

#### 3.2.1.2 African bollworm

Chemicals approved for bollworm control are aimed at younger larvae because older larvae are less susceptible to chemical treatment. When three to four larvae are present per meter row in dryland conditions, chemical control may be used. Spraying can be considered in irrigated fields when six to seven larvae are present, due to the higher seeding rate.

### 3.2.1.3 Leaf miner fly

Rotating wheat with oats to break the pest's life cycle is the most effective way to control leaf miner fly where severe problems occur annually. Leaf miners can live in the soil for up to a year. When oats are planted after maize, the hatching leaf miners will most likely look for a field of wheat, and with no pupae produced in the soil, the problem can be reduced or eradicated.

## 3.2.2 Diseases

### 3.2.2.1 Rusts

Growing resistant cultivars and applying fungicides are the two most common methods of wheat rust control. Rust-resistant cultivars offer an efficient and environmentally friendly method of rust control. As a result, rust resistance has been a key component of the ARC-Small Grain wheat breeding objectives (ARC-SG). Several resistant cultivars have been developed in recent years. The resistance status of commercially available cultivars is tested with relevant rust races once a year, and the results are published by ARC-SG in the *Small Grain Production Guideline* on a regular basis. Farmers can use this information to decide which cultivar to grow in a particular production area.

Using resistant cultivars to control wheat rusts presents some challenges. One of the most significant limitations of resistant cultivars is that rust-causing fungi frequently acquire new virulence to overcome resistance in existing

cultivars. The majority of virulent races emerge locally as a result of genetic mutation in the existing rust population. New rust races can be introduced into South Africa from other countries via windborne spores or, more likely, by adhering to travelers' clothing. As a result, ARC-SGI has been conducting annual rust surveys (monitoring) to detect potentially dangerous rust races in a timely manner, and the results have been used to continuously improve wheat rust resistance in cultivars that have been released on a regular basis.

Fungicides can be used to control wheat rusts when resistant cultivars are not available or when resistance in existing cultivars breaks down due to the emergence of new races. Excessive use of fungicides, on the other hand, is harmful to the environment and may pose a health risk to farmers and farmworkers. It may also increase the likelihood of the emergence of fungicide-resistant rust strains, rendering the chemicals ineffective. As a result, fungicides should only be used on susceptible cultivars where rusts are likely to cause significant yield loss. Fungicides should not be used on resistant wheat cultivars.

The following manual can be consulted for use to control wheat stem rust (De Villiers et al, 2019).

## 3.2.3 Weeds

Weeds can be controlled using both [IPM](#) and [CA](#) strategies. The IPM manual can be found at <https://www.agriseta.co.za/downloads/LearningMaterial/116301LG.pdf>.

## 4 CONCLUSION

Climate change's effects on crop production must be thoroughly understood before effective mitigation strategies can be implemented. In order for [mitigation](#) strategies to work, it is also necessary to practice the timing of cultural practices in wheat production.

One of the strategies to minimise the application of chemicals such as pesticides and fungicides while also protecting the environment is to develop high yielding and widely adapted wheat genotypes with increased water-use efficiency, heat tolerance, and resistance to major diseases and pests (Tadesse et al, 2018). The use of these well-adapted varieties; a diverse range of crop species and varieties grown in associations, intercroops, or rotations; pest control through integrated pest management; and the use of conservation agriculture and sustainable mechanization can maintain healthy soils and efficiently manage water to achieve the highest possible productivity per unit of input.

When one cultivar is affected and the other is not, cultivating a diverse suite of cultivars in associations, intercroops, and/or rotations can be beneficial ([FAO](#), 2011).

It is critical to understand the correct operations to perform on the small grain field, the correct time to perform them, the correct way to perform them, as well as why the operations are performed, why they are performed at a specific time, and why they are performed in a specific manner. Planting rates and timing are two examples of these. Knowing planting dates and seeding rates determines the best combinations, and shifting planting

dates to accommodate new temperature and precipitation patterns is the simplest and least expensive adaptation. It is critical to plant wheat cultivars at the appropriate time to ensure flowering occurs during the optimal window for maximum grain yield potential. The optimal flowering window is determined by a balance of water used during canopy development, grain formation, and grain-filling phases, as well as the frequency and severity of frosts decreasing. Crops that flower too early are more likely to be frost damaged, whereas crops that flower too late are more likely to be subjected to high temperatures and water deficits, which can limit grain formation and grain filling.

The wheat crop can also be planted earlier to avoid terminal heat, but this increases the risk of being exposed to heavy rain (which is largely unpredictable). As a result, the timing of sowing as well as the wheat production systems must be adjusted. Because of the time saved with direct seeding, [CA](#) allows for an earlier wheat sowing date when compared to the traditional tillage-based wheat production system.

In irrigated systems, increasing irrigation efficiency (e.g., through deficit irrigation, precise water applications, high-efficiency pumps), reducing water losses, and improving water allocation and demand management optimises yields per volume of water applied, reduces greenhouse gas emissions, and results in energy efficiency gains, primarily in fuel use. In addition, introducing supplemental or deficit irrigation in rain-fed conditions is an effective way of maintaining or increasing grain yield in dryland conditions.



## 5 REFERENCES AND RESOURCES

- Al-Hadeethi I, Li Y, Odhafa AKH, Al-Hadeethi H, Seneweera S, Lam SK (2019). Assessment of grain quality in terms of functional group response to elevated [CO<sub>2</sub>], water, and nitrogen using a meta-analysis: Grain protein, zinc, and iron under future climate. *Ecol Evol.* 2019;9:7425–7437. <https://doi.org/10.1002/ece3.5210>).
- Asseng S, Ewert F, Martre P, Rotter RP & Lobell DB (2014). Rising temperatures reduce global wheat production. *Nat. Climat. Change.* 5:143-47.
- Bunce JA (1993). Growth, survival, competition, and canopy carbon dioxide and water vapor exchange of first year alfalfa at an elevated CO<sub>2</sub> concentration. *Photosynthetica.* 29: 557-565.
- Bunce JA & Ziska LH (2000). Crop ecosystem responses to climatic change: crop/weed interactions. In: Reddy, K.R., Hodges, F. (Eds.), *Climate Change and Global Crop Productivity*. CABI, New York, pp. 333-348.
- Butler EE & Huybers P (2015). Variations in the sensitivity of US maize yield to extreme temperatures by region and growth phase. *Environ. Res. Lett.* 10:034009.
- Carlson RE (1990). Heat stress, plant-available soil moisture, and corn yields in Iowa: a short- and longterm view. *J. Prod. Agric.* 3: 293-297.
- Chandra SK, Kumar M, Singh SK & Kumari A (2013). Impact of Climate Change on Soil Health and Fertility. *Proceedings: National Seminar on Climate Change and Indian Horticulture Exploring Adaptation and Mitigation Strategies for Expeditious Resilience.*
- CSIRO & Bureau of Meteorology (2015). *Climate Change in Australia Information for Australia's Natural Resource Management Regions: Technical Report*. CSIRO and Bureau of Meteorology, Australia.
- De Villiers C, Hatting J, Terefe T & Laing M (2019). A novel study: Biological suppression of wheat stem rust. *On Farm Level Integrated Pest Management. SA Grain.* Pp 78-79. <http://www.arc.agric.za/Agricultural%20Sector%20News/A%20novel%20study%20-%20Biological%20suppression%20of%20wheat%20stem%20rust.pdf>
- Easterling WE, Aggarwal PK, Batima P, Brander KM, Erda L, Howden SM, Kirilenko A, Morton J, Soussana J-F, Schmidhuber J & Tubiello FN (2007). Food, fibre and Forest products. Contribution of working group II to the fourth assessment report of IPCC.
- FAO (2011). *Save and grow, a policymaker's guide to the sustainable intensification of smallholder crop production*. Rome.

- Flato GJ, Marotzke B, Abiodun P, Braconnot SC, Chou W, Collins P, Cox F, Driouech S, Emori V, Eyring C, Forest P, Gleckler E, Guilyardi C, Jakob V, Kattsov C, & Rummukainen M (2013). Evaluation of climate models. In Climate Change 2013: The Physical Science Basis. Contribution of Working Group I to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change. T.F. Stocker, D. Qin, G.-K. Plattner, M. Tignor, S.K. Allen, J. Doschung, A. Nauels, Y. Xia, V. Bex, and P.M. Midgley, Eds. Cambridge University Press, pp. 741-882, doi:10.1017/CBO9781107415324.020.
- Girvetz EH, Zganjar C (2014) Dissecting indices of aridity for assessing the impacts of global climate change. *Climate Change*. 126: 469-483.
- Girvetz E, Ramirez-Villegas J, Claessens L, Lamanna C, Navarro-Racines C, Nowak A, Thornton P & Rosenstock TS (2019). Chapter 2 Future Climate Projections in Africa: Where Are We Headed?
- Prinsloo G (2018). Wheat leaf miner does not like oats. *Wheat Focus*. March/April 2018. Page 9-11.
- Prinsloo G (2018). Migration patterns of aphids explained. *SAGrain*. June 2018. Page 68-69.
- Prinsloo G (2017). The big 3 insect pests on wheat. *Wheat Focus* September/October Issue Page 18-19.
- GrainSA (2015). Introduction to Conservation Agriculture. GrainSA Conservation Farmer Innovation Programme.
- Grube A, Donaldson D, Kiely T, Wu L (2011). Pesticide industry sales and usage: 2006 and 2007 market estimates. EPA's Biological and Economic Analysis Division, Office of Pesticide Programs, Office of Chemical Safety and Pollution Prevention, U.S. EPA. [http://www.epa.gov/opp00001/pestsales/07pestsales/market\\_estimates2007.pdf](http://www.epa.gov/opp00001/pestsales/07pestsales/market_estimates2007.pdf) (accessed 24.02.2021).
- Kibblewhite MG, Ritz K & Swift MJ (2008). Soil health in agricultural systems. *Philosophical Transactions of the Royal Society B* 363: 685-701. doi:10.1098/rstb.2007.2178.
- Linden PJ & Hanson CE (2007). *Climate Change 2007: Impacts, Adaptation and Vulnerability*, Cambridge University Press Cambridge, pp. 273-313.
- McErlich AF & Boydston RA (2013). Current state of weed management in organic and conventional cropping systems. Publications from USDA-ARS/UNL Faculty. Paper 1387.
- Myers SS, Zanobetti A, Kloog I, Huybers P, Leakey ADB, Arnold JB, Carlisle E, Dietterich LH, Fitzgerald G, Hasegawa T, Holbrook NM, Nelson RL, Ottman MJ, Raboy V, Sakai H, Sartor KA, Schwartz J, Seneweera S, Tausz M & Usui Y (2014) . Increasing CO<sub>2</sub> threatens human nutrition. *Macmillan Publishers Limited* doi:10.1038/nature13179.

- Myers SS, Smith MR, Guth S, Golden CD, Vaitla B, Mueller ND, Alan D, Dangour AD & Huybers P (2017). Climate Change and Global Food Systems: Potential Impacts on Food Security and Undernutrition. *Annu. Rev. Public Health.* 38: 259-77. <https://doi.org/10.1146/annurev-publhealth-031816-044356>.
- Parmesan C, Yohe G (2003). A globally coherent fingerprint of climate change impacts across natural systems. *Nature.* 421: 37-42.
- S´anchez B, Rasmussen A & Porter JR (2014). Temperatures and the growth and development of maize and rice: a review. *Glob. Change Biol.* 20: 408-417.
- Schlenker W, Roberts MJ (2009). Nonlinear temperature effects indicate severe damages to U.S. crop yields under climate change. *PNAS* 106:15594-15598.
- Smith MR, Myers SS (2018). Impact of anthropogenic CO<sub>2</sub> emissions on global human nutrition. *Nature Clim Change.* 8: 834-839. <https://doi.org/10.1038/s41558-018-0253-3>.
- Southern African Grain Laboratory (SAGL) (2018). South African Wheat crop quality report 2017/2018 season. Pretoria: SAGL.
- Southern African Grain Laboratory (SAGL) (2019). South African Wheat crop quality report 2018/2019 season. Pretoria: SAGL.
- Tadesse W, Bishaw Z & Assefa S (2018). Wheat production and breeding in Sub-Saharan Africa - Challenges and opportunities in the face of climate change. *International Journal of Climate Change Strategies and Management* 11 (5): 696-715 DOI 10.1108/IJCCSM-02-2018-0015.
- Trnka M, Feng S, Semenov MA, Olesen JE, Kersebaum KC, Rötter RP, Semerádová D, Klem K, Huang W, Ruiz-Ramos M, Hlavinka P, Meitner J, Balek J, Havlík P & Büntgen U (2019). Mitigation efforts will not fully alleviate the increase in water scarcity occurrence probability in wheat-producing areas. *Science Advances* DOI: 10.1126/sciadv.aau240
- Ziska LH, Teasdale JR & Bunce JA (1999). Future atmospheric carbon dioxide may increase tolerance to glyphosate. *Weed Sci.* 47: 608-615.
- Ziska LH (2008). Rising atmospheric carbon dioxide and plant biology: the overlooked paradigm. *DNA Cell Biol.* 27: 165-172.

## LIST OF FIGURES

|           |                                                                                                                                                        |     |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| Figure 1  | Map showing both dryland wheat production areas in winter rainfall areas of Western Cape Province.                                                     | 188 |
| Figure 2  | Map showing both dryland and irrigated wheat production areas in summer rainfall areas of SA.                                                          | 190 |
| Figure 3  | Projection of the possible shifts in the wheat production areas for the summer rainfall areas where spring wheat is planted in the easterly provinces. | 194 |
| Figure 4  | Projection of the possible shifts in the wheat production areas under irrigation in the summer rainfall areas.                                         | 195 |
| Figure 5  | Projection of the possible shifts in the wheat production areas for Western Cape winter rainfall area.                                                 | 195 |
| Figure 6  | Russian wheat aphid.                                                                                                                                   | 197 |
| Figure 7  | Oat aphid.                                                                                                                                             | 199 |
| Figure 8  | Rose grain aphid.                                                                                                                                      | 199 |
| Figure 9  | Brown ear aphid.                                                                                                                                       | 200 |
| Figure 10 | African bollworm and the damage it causes.                                                                                                             | 200 |
| Figure 11 | Leaf rust.                                                                                                                                             | 201 |
| Figure 12 | Stem rust.                                                                                                                                             | 202 |
| Figure 13 | Stripe rust.                                                                                                                                           | 202 |
| Figure 14 | Barley yellow dwarf virus infected wheat leaves.                                                                                                       | 204 |
| Figure 15 | Wild buckwheat seedling.                                                                                                                               | 205 |
| Figure 16 | Wild buckwheat in an oats field.                                                                                                                       | 205 |
| Figure 17 | Young Chenopodium plant.                                                                                                                               | 206 |
| Figure 18 | Chenopodium seeds.                                                                                                                                     | 206 |
| Figure 19 | Avena plant.                                                                                                                                           | 206 |
| Figure 20 | Avena spikelets.                                                                                                                                       | 207 |
| Figure 21 | Healthy loose soil with good structure versus unhealthy cloddy soil with poor structure.                                                               | 208 |
| Figure 22 | Earthworms in healthy soil.                                                                                                                            | 209 |
| Figure 23 | Wheat seedlings growing on healthy soil with well decomposed humus.                                                                                    | 210 |
| Figure 24 | Minimum tillage planting.                                                                                                                              | 212 |

|           |                                                                                                                       |     |
|-----------|-----------------------------------------------------------------------------------------------------------------------|-----|
| Figure 25 | Soil covered with crop residue.                                                                                       | 213 |
| Figure 26 | Soil in between wheat rows covered with crop residue.                                                                 | 214 |
| Figure 27 | Soil covered with cover crop between rows of wheat crop.                                                              | 214 |
| Figure 28 | Crop diversification by alternating crops at the same time.                                                           | 215 |
| Figure 29 | Wheat mix cropped with clover.                                                                                        | 216 |
| Figure 30 | Crop rotation, starting from top left, lucerne - wheat - barley<br>- lupin - wheat - barley - canola - lucerne again. | 217 |

## LIST OF TABLES

|         |                                                                                                                                  |     |
|---------|----------------------------------------------------------------------------------------------------------------------------------|-----|
| Table 1 | Wheat production overview over three seasons.                                                                                    | 189 |
| Table 2 | Resistance(R)/Susceptibility (S) of wheat germplasm<br>with resistant genes against five South African biotypes (RWASA1-RWASA5). | 198 |
| Table 3 | Environmental conditions required for the wheat rusts.                                                                           | 203 |
| Table 4 | Effects of different management practices on gains and losses<br>of organic matter, beneficial organisms and pests.              | 211 |